Random Variables: Variance

Jericho Lawson

MATH 241

Week 7, Lecture #2 — Ch. 4.6

Reminders!

- Homework #5 due Friday at 11:59pm
  - No HW assignment over break!

Variance

The variance of a random variable X is:

$$E[(X-\mu)^2] = \sum_x (x-\mu)^2 p_X(x)$$

We can also write this in another way.

Variance

$$
\begin{aligned}
Var(X) &= E[(X-\mu)^2] \\
&= E(X^2 - 2X\mu + \mu^2) \\
&= E(X^2) - E(2X\mu) + E(\mu^2) \\
&= E(X^2) - 2\mu E(X) + \mu^2 \\
&= E(X^2) - 2\mu^2 + \mu^2 \\
&= E(X^2) - \mu^2 \\
&= E(X^2) - E^2(X)
\end{aligned}
$$

Variance

Much easier to calculate the variance of X uising the second formula:

$$Var(X) = E(X^2) - E^2(X)$$

- Uses first two moments: $E(X)$ and $E(X^2)$
- Measures spread of random variable
- Denoted as σ_X^2
  - σ_X is the standard deviation of RV X

Example: Dice Rolls

Suppose X is the number we roll for a single die. Find the **variance**.

Example: Dice Rolls

Suppose X is the number we roll for a single die. Find the **variance**.

We now need to find (1) $E(X)$ and (2) $E(X^2)$. First, we start with the support of X and $P(X = x)$:

$X = x$	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Example: Dice Rolls

Suppose X is the number we roll for a single die. Find the **variance**.

Then, we find $E(X)$:

$$\begin{aligned} E(X) &= \sum_x x\, p_X(x) \\ &= 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) \\ &= \frac{7}{2} \end{aligned}$$

Example: Dice Rolls

Suppose X is the number we roll for a single die. Find the **variance**.

Then, we find $E(X^2)$:

$$\begin{aligned} E(X^2) &= \sum_x x^2 \, p_X(x) \\ &= 1\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 9\left(\frac{1}{6}\right) + 16\left(\frac{1}{6}\right) + 25\left(\frac{1}{6}\right) + 36\left(\frac{1}{6}\right) \\ &= \frac{91}{6} \end{aligned}$$

Example: Dice Rolls

Suppose X is the number we roll for a single die. Find the **variance**.

Then, we find $Var(X)$:

$$\begin{aligned} Var(X) &= E(X^2) - E^2(X) \\ &= \frac{91}{6} - \left(\frac{7}{2}\right)^2 \\ &\approx 2.9167 \end{aligned}$$

Activity: Finding a Variance

Let X be the number of club members who attend a monthly meeting, where the club has 6 members. Suppose we have the following probability distribution:

- $P(X = 0) = 0.02$
- $P(X = 1) = 0.07$
- $P(X = 2) = 0.11$
- $P(X = 3) = 0.2$
- $P(X = 4) = 0.23$
- $P(X = 5) = 0.29$
- $P(X = 6) = 0.08$

Find the variance of X.

Properties of Variance

1. Linearity

$$Var(aX + b) = a^2 Var(X)$$

   - On your own: Show this holds using expectations.

2. 2 RVS (not independent)

$$Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

3. 2 RVs (independent)

$$Var(X + Y) = Var(X) + Var(Y)$$

4. n RVs (independent)

$$Var\left(\sum_i X_i\right) = \sum_i Var(X_i)$$

Activity: Binomial Distribution, Revisited

Say we flip four weighted coins independently and are curious about X, the number of heads I get in these four flips. The chance of getting heads is 60%. Find the expectation of X.

1. Show that $Var(X)$ for any binomial RV is $np(1-p)$.
   - Hint: Explore a Bernoulli RV first, then connect pieces to a Binomial RV.
2. Find the variance of X defined above:
   - By hand using the general definition of a variance, and
   - Using $np(1-p)$.